2026 MIDDLE SCHOOL STEM CAMPS

Master Curriculum and Operations Guide

Two camps, 24 primary missions and 8 backups, built on one engineering-design loop: predict, build or solve, test, score, redesign, defend.

EXECUTIVE SUMMARY

The program runs as two distinct STEM seasons for 15 to 25 middle school students in 4 to 6 teams. From Trees to Tech treats nature as the engineer: trees, leaves, soil, water, ecosystems, sensors, and climate resilience. PY-STEM is broad applied STEM: biomedical engineering, waves, optics, mechanics, materials, accessibility design, and algorithms, with light tie-ins to the Science History Institute and the Charles Library BookBot.

Every primary activity has a measurable win condition and a 100-point rubric. Competition is built into the science task itself, not added on top. The two camps share no core mechanic, build type, or scoring loop.

FROM TREES TO TECH: 12 PRIMARY ACTIVITIES

ID	ACTIVITY	CORE IDEA AND WIN
TTT-01	MudWatt Bioelectric League (Biology + Energy)	Highest combined score from peak reading, daily data quality, controlled-variable design, redesign notes, and final explanation.
TTT-02	Forest Sensor Sprint (Field Science)	Best evidence-based sensor recommendation with accurate measurements and efficient route planning.
TTT-03	Seed Dispersal Derby (Biomimicry)	Best combined distance, hang time, target landing, biomimicry explanation, and redesign improvement.
TTT-04	Xylem Pipeline Relay (Biomimicry)	Most water delivered with speed, low spill, and strong explanation.
TTT-05	Greenhouse Climate Controller (Ecology)	Best plant-zone-control match supported by tour evidence.
TTT-06	Pinecone Weather Machine (Biomimicry)	Most accurate, readable, and fast response after spray-and-dry trials.
TTT-07	Pollinator Network Draft and Build (Ecology)	Highest pollinator support across spring, summer, and fall using native and clumping logic.
TTT-08	Arboretum Eco-Quest (Field Science)	Most accurate checkpoint solutions with safe field conduct.
TTT-09	Minecraft Tree World Resilience Cup (Ecology)	Best climate-resilient design with accurate nature-based strategies.

ID	ACTIVITY	CORE IDEA AND WIN
TTT-10	Tree Ring Climate Detective (Data + Climate)	Most accurate climate-event inferences with claim-evidence-reasoning.
TTT-11	Lotus Leaf Surface Sprint (Materials)	Best clean runoff and least residue after one redesign.
TTT-12	Leaf Stomata Microscope Detective (Field Science)	Most accurate detective ranking with clean data and strong evidence.

PY-STEM: 12 PRIMARY ACTIVITIES

ID	ACTIVITY	CORE IDEA AND WIN
PYS-01	Magnetic Capsule Maze Cup (Biomedical)	Fastest clean run with low penalties and a clear magnetic-control explanation.
PYS-02	Oobleck Armor Arena (Materials)	Best target protection with least material and cleanest reset.
PYS-03	Cardboard Automata Arcade (Mechanics)	Most reliable motion with useful mechanism and clear explanation.
PYS-04	Stethoscope Sprint and Recovery (Biomedical)	Best sound quality plus accurate, respectful data collection.
PYS-05	Reaction Time Combine (Biomedical)	Best median plus most improvement using clean data.
PYS-06	SONAR Slinky Showdown (Waves + Sound)	Most correct wave predictions and station solutions.
PYS-07	Pinhole Precision Challenge (Optics + Light)	Best image clarity and concept explanation after one aperture redesign.
PYS-08	Low-Ropes Force Map Relay (Mechanics)	Most accurate predictions and clear debrief connections.
PYS-09	Hovercraft Hockey Hackathon (Mechanics)	Best glide plus control in a gym-safe tournament.
PYS-10	Spectra Sleuth Showdown (Optics + Light)	Most correct spectrum matches and strongest safe fireworks-science explanation.
PYS-11	BookBot Bin Logic Challenge (Computing)	Most correct retrievals with few traffic conflicts and clear algorithm defense.
PYS-12	Accessibility Ramp Rescue Lab (Design)	Best safe ramp meeting slope, load, portability, and client criteria.

BACKUP ACTIVITIES

Held ready for weather, trip delays, or materials failure. Same loop, lighter setup.

ID	ACTIVITY	CORE IDEA AND WIN
TTB-01	Root Grip Erosion Rescue (Field Science)	Most soil retained after standard pour.
TTB-02	Tree Height Triangulation Shootout (Field Science)	Closest to instructor reference height.
TTB-03	Urban Heat Shade Dash (Data + Climate)	Best data-backed cool route.
TTB-04	Photosynthesis Float-Off Playoffs (Field Science)	Fastest half-float time with controlled variables.
PYB-01	Pencil Pressure Safety Lab (Mechanics)	Largest area protected with clear data.
PYB-02	Domino Neuron Relay (Biomedical)	Fastest reliable signal with best explanation.
PYB-03	Pulley Rescue Relay (Mechanics)	Best lift with correct setup and explanation.
PYB-04	Barcode Checksum Rescue (Computing)	Most bad codes detected with few false alarms.

THE SHARED LEARNING LOOP

Every activity moves teams through the same cycle, which mirrors the engineering design process and the practices in NGSS MS-ETS1.

- Predict: teams commit to a prediction and a reason before any materials move.
- Build or solve: teams construct or work the challenge using approved materials.
- Test: teams run a standard test and record a baseline result.
- Score: the 100-point rubric rewards design, data quality, teamwork, and explanation.
- Redesign: teams change one variable, predict its effect, and re-test.
- Defend: teams justify their final claim with specific evidence.

SAFETY OVERVIEW

No student open flames, dry ice, high voltage, hazardous chemicals, pressure vessels, dangerous projectiles, major staining, or cut-open glow sticks are used anywhere in the program.

- All electrical work is at battery voltage. MudWatt cells are sealed and handled with gloves.
- Magnets are counted and controlled. Oobleck and water activities use trays and immediate cleanup.
- Pinhole optics include a no-direct-sun-view rule. Field trips follow host rules first.
- PPE is required where splash, soil, oobleck, plant peel, or cleanup risk exists: goggles, nitrile gloves, handwashing.
- Allergy notes appear in each activity guide. The full allergy roster is confirmed before camp.

Detailed daily tasks live in the Staff Setup, Prep, and Safety Checklist.

OPERATIONS AND ROLES

- Core rule: bins packed before instruction starts, score sheets ready before testing starts, PPE visible before materials are distributed.
- Each team rotates roles: captain, safety lead, builder, tester, data lead, presenter.
- Best 9 of 12 primary scores count toward the final leaderboard so a single rough activity does not sink a team.
- Backups are staged and ready every day, not improvised on the spot.

MATERIALS AND PACKING BINS

Each activity has a labeled bin. Procurement, reusable-versus-consumable status, and budget scenarios are tracked in the Amazon Procurement Workbook. Unverified prices and ASINs must be checked on the purchase date.

PRODUCTION NOTES

- Leaf chromatography and tie-dye were removed; Leaf Stomata Microscope Detective is used instead.
- BookBot is framed as automated storage and retrieval, bin addressing, and routing logic, not conveyor throughput.
- PY-STEM stays broad, with only selected anchor tie-ins to the Science History Institute and the BookBot.
- The two camps were audited against each other and against the legacy no-repeat list; no banned activity is reused.